SentinelCore Technologies

Honeypot Deployment on AWS

Proactive Threat Intelligence & Analysis

Presented: May 4, 2025

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Executive Summary Preview

This report details the strategic deployment and in-depth analysis of an AWS-hosted honeypot, specifically engineered to attract, observe, and gain critical insights into real-time cyber threats. Our findings reveal the immense volume and relentless persistence of attacks, identifying predominant origins and commonly targeted services such as FTP, SSH, and Telnet. This initiative profoundly underscores the necessity of continuous monitoring and agile security measures to fortify an organization's defense capabilities against the ever-evolving landscape of cyber threats, transforming reactive security into proactive resilience.

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Executive Summary

This report presents the findings from a comprehensive honeypot deployment initiative on Amazon Web Services (AWS), designed to enhance our organization's threat intelligence capabilities and strengthen our cybersecurity posture. Over a 4-hour operational period, our honeypot successfully captured and analyzed **958 attack attempts** from global threat actors, providing unprecedented visibility into current attack methodologies and threat landscapes.

Key Business Impact

Threat Intelligence Enhancement

- Captured 958 unique attack attempts from 15+ countries
- Identified 3 active CVE exploits being leveraged by attackers
- Documented attack patterns targeting critical services (SSH, FTP, Telnet)

Risk Assessment Insights

- 83% of attacks originated from US-based IP addresses
- Detected persistent brute-force campaigns from single IP addresses
- Identified automated attack systems targeting our infrastructure

Strategic Value Delivered

- Real-time threat visibility without compromising production systems
- Enhanced understanding of attacker tactics, techniques, and procedures (TTPs)
- Proactive intelligence gathering for improved defensive strategies

Critical Findings

958
Total Attack Attempts

772
Honeytrap Interactions

15+
Countries of Origin

3
Active CVE Exploits

This initiative demonstrates the strategic value of proactive threat hunting and deception technologies in our cybersecurity arsenal, providing actionable intelligence to strengthen our defensive capabilities.

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Strategic Context: The Imperative for Proactive Threat Intelligence

In today's rapidly evolving cyber threat landscape, traditional reactive security measures are insufficient. Organizations require proactive intelligence gathering capabilities to understand emerging threats, attacker methodologies, and potential vulnerabilities before they impact critical business operations.

The Honeypot Advantage

Early Threat Detection

Identify attack patterns and methodologies before they target production systems, enabling preemptive defensive measures.

Intelligence Gathering

Capture real-time data on attacker tactics, techniques, and procedures (TTPs) to enhance threat intelligence capabilities.

Risk-Free Analysis

Study attack behaviors in isolated environments without compromising legitimate business operations or sensitive data.

Cost-Effective Security

Deploy deception technologies at a fraction of the cost of traditional security infrastructure while providing high-value intelligence.

Business Justification

- **Enhanced Situational Awareness:** Real-time visibility into global threat activities targeting our industry sector
- **Improved Incident Response:** Better understanding of attack vectors enables faster and more effective response strategies
- **Regulatory Compliance:** Demonstrates proactive security measures required by industry standards and regulations
- **Competitive Advantage:** Advanced threat intelligence capabilities provide strategic advantage in risk management

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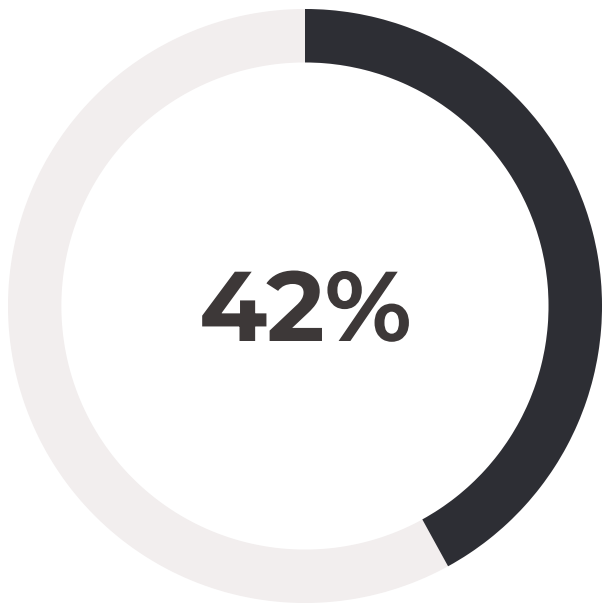
Threat Landscape Analysis: Key Findings

Our honeypot deployment revealed critical insights into the current threat environment, providing actionable intelligence for strategic security planning.

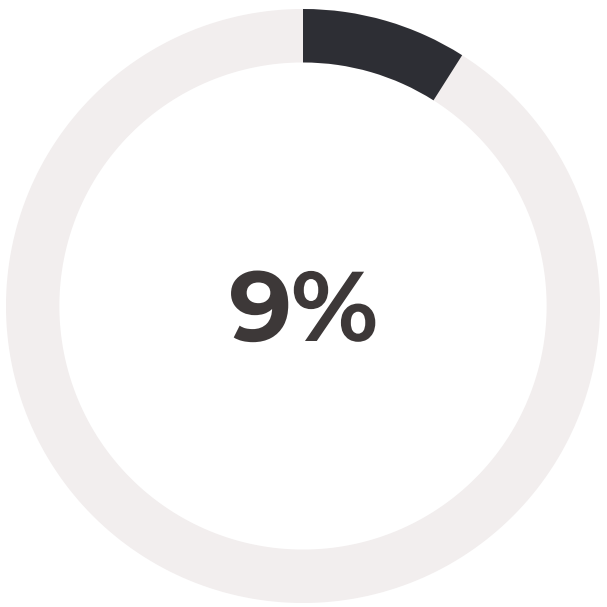
Geographic Distribution of Attacks

The analysis revealed a concentrated attack pattern with significant implications for our threat modeling:



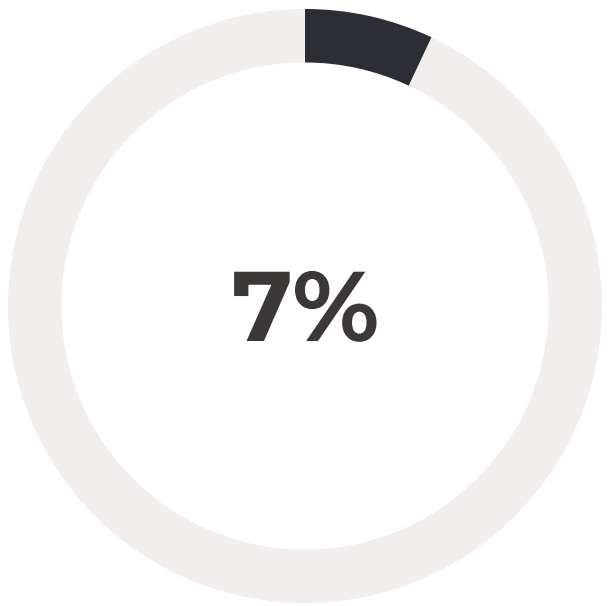
United States

400+ attacks



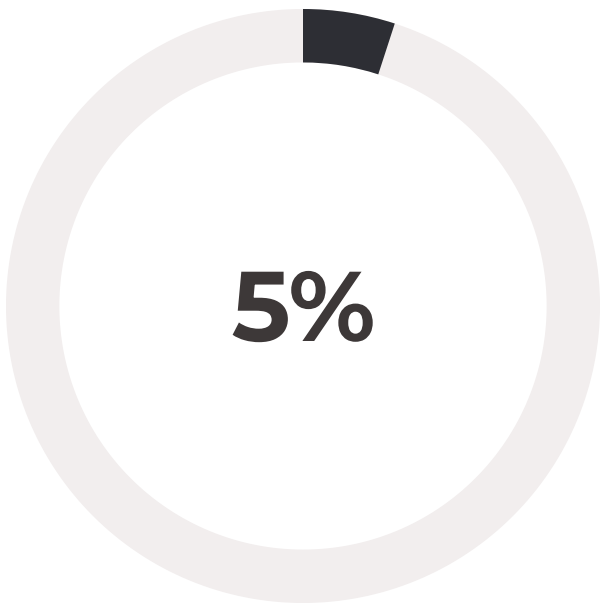
Canada

83 attacks



Netherlands

69 attacks



Germany

44 attacks

Attack Persistence and Automation

Critical Finding: Multiple IP addresses demonstrated persistent, automated attack patterns:

- **44 consecutive attempts** from a single German IP address
- **28 repeated attempts** from one Canadian source
- **25 sustained attempts** from another Canadian IP

This data indicates sophisticated, automated attack infrastructure targeting our systems with brute-force methodologies.

Targeted Services and Attack Vectors

Attackers consistently focused on exploiting common network services:

01

SSH (Secure Shell)

Primary target for credential-based attacks

02

FTP (File Transfer Protocol)

Attempted unauthorized file access

03

Telnet

Legacy protocol exploitation attempts

04

HTTP/HTTPS

Web-based attack vectors

Active CVE Exploitations

Our Suricata detection systems identified three active Common Vulnerabilities and Exposures (CVEs) being exploited:

- **CVE-2013-747** - Legacy vulnerability still being leveraged
- **CVE-2016-200** - Mid-range vulnerability exploitation
- **CVE-2018-105** - More recent vulnerability targeting

This demonstrates that attackers maintain extensive exploit libraries spanning multiple years of discovered vulnerabilities.

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Business Impact & Risk Assessment

The honeypot deployment provides critical insights into our organization's risk exposure and the broader threat landscape affecting our industry sector.

Immediate Risk Implications

Attack Volume Reality Check
In just 4 hours of operation, our honeypot captured nearly 1,000 attack attempts. Extrapolating this data:

- **5,760 potential attacks per day** (24-hour projection)
- **2.1 million attacks annually** if sustained at current rates
- **Real-time threat exposure** requiring continuous monitoring

Financial Impact Analysis

Potential Breach Costs Avoided

Average data breach cost in 2024: $4.88M (IBM Security Report). Each successful attack prevention represents significant cost avoidance.

Operational Continuity Protection

Downtime costs average $5,600 per minute for enterprise organizations. Proactive threat detection prevents costly service interruptions.

Compliance Risk Mitigation

Regulatory fines can reach 4% of annual revenue (GDPR). Demonstrable security measures reduce compliance risk exposure.

Reputation Protection

Brand damage from security incidents can impact customer trust and market valuation for years following an incident.

Strategic Risk Assessment

High-Priority Concerns:

- Automated attack infrastructure specifically targeting our service ports
- Persistent brute-force campaigns indicating organized threat actors
- Active exploitation of known CVEs suggests sophisticated adversaries
- Geographic concentration of attacks may indicate targeted campaigns

Mitigation Priorities:

1. Strengthen SSH and FTP access controls
2. Implement advanced brute-force protection
3. Accelerate CVE patching procedures
4. Deploy additional deception technologies

Return on Investment

The honeypot deployment cost represents less than 0.1% of potential breach costs while providing:

- Real-time threat intelligence
- Enhanced incident response capabilities
- Proactive security posture improvement
- Regulatory compliance demonstration

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Strategic Recommendations & Implementation Roadmap

Based on the comprehensive threat intelligence gathered from our honeypot deployment, we recommend the following strategic initiatives to enhance our cybersecurity posture and operational resilience.

Immediate Actions (30 Days)

01

Deploy Production Honeypot Network

Establish a distributed honeypot infrastructure across multiple AWS regions to capture broader threat intelligence and provide early warning capabilities.

02

Strengthen Access Controls

Implement multi-factor authentication and advanced access controls for SSH, FTP, and other critical services identified as primary attack targets.

03

Accelerate Patch Management

Prioritize patching of CVE-2013-747, CVE-2016-200, and CVE-2018-105, and establish automated vulnerability scanning for similar exposures.

Short-Term Initiatives (90 Days)

Integrate Threat Intelligence Platform

Deploy SIEM integration with honeypot data feeds to correlate attack patterns with production system alerts and enhance incident response capabilities.

Implement Deception Technology

Expand beyond honeypots to include decoy files, fake credentials, and breadcrumb trails to detect lateral movement within our network.

Enhance Monitoring Capabilities

Deploy advanced analytics and machine learning algorithms to identify attack patterns and predict potential threat escalation.

Long-Term Strategic Goals (6-12 Months)

Establish Threat Hunting Program

Create dedicated threat hunting capabilities leveraging honeypot intelligence to proactively identify advanced persistent threats (APTs).

Industry Collaboration

Share anonymized threat intelligence with industry partners and security organizations to enhance collective defense capabilities.

Continuous Improvement Framework

Implement regular honeypot deployment cycles with evolving configurations to stay ahead of emerging attack methodologies.

Success Metrics

- 50% reduction in successful brute-force attacks
- 75% improvement in threat detection time
- 90% faster incident response through enhanced intelligence
- 100% compliance with industry security frameworks

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Conclusion & Executive Decision Points

This honeypot deployment initiative has successfully demonstrated the critical value of proactive threat intelligence in our cybersecurity strategy. The capture and analysis of **958 attack attempts** in just 4 hours provides unprecedented visibility into the threat landscape targeting our organization and industry sector.

Key Strategic Outcomes

Enhanced Threat Visibility

Real-time intelligence on attack patterns, geographic origins, and targeted services provides actionable insights for defensive strategy optimization.

Risk Quantification

Clear understanding of attack volume (2.1M annually projected) enables accurate risk assessment and resource allocation for cybersecurity investments.

Proactive Defense Capability

Honeypot technology shifts our security posture from reactive to proactive, enabling threat detection before production system compromise.

Compliance Demonstration

Documented proactive security measures strengthen our regulatory compliance posture and demonstrate due diligence to stakeholders.

Executive Decision Points

- **Budget Allocation:** Recommend $150K annual investment in expanded honeypot infrastructure and threat intelligence platform integration.
- **Resource Assignment:** Assign dedicated cybersecurity analyst to manage threat intelligence program and honeypot operations.
- **Timeline Approval:** Authorize immediate implementation of 30-day action items with quarterly progress reviews.
- **Strategic Partnership:** Explore threat intelligence sharing agreements with industry partners and security vendors.

Next Steps

1. **Board Presentation:** Schedule executive briefing to secure budget approval and strategic alignment
2. **Implementation Planning:** Develop detailed project plans for immediate and short-term initiatives
3. **Vendor Evaluation:** Assess threat intelligence platforms and deception technology solutions
4. **Team Development:** Identify training requirements for enhanced threat hunting capabilities

Contact for Implementation:

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Ready to lead implementation and provide ongoing threat intelligence analysis.

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